

Smart Meter Big Data Processing — Experimental Summary

PHASE 2 EXPERIMENTAL RESULTS

DATA VOLUME EXPERIMENT

- Tested volumes: 1M, 5M, 10M records
- All runs completed successfully
- 10M throughput: 931,671.81 records/s
- 10M execution time: 10.733 s

REPLAY RATE EXPERIMENT

- Requested rates: 100, 500, 1,000 records/s
- All runs delivered 10,000 / 10,000 records
- Highest measured producer rate: 745.80 records/s
- Highest measured streaming rate: 712.45 records/s
- Record delivery: 100% for every tested run

FULL-DATA PRODUCTION

- Anomaly dataset: 167,811,461 valid records
- Forecasting dataset: 19,862 hourly records
- Distinct households: 5,561
- Output format: Parquet
- Temporal split: chronological
- Future leakage: not permitted